University of Arizona Cyberinfrastructure and Research Support 2026

Introduction

The University of Arizona (UA/UArizona) is the Arizona Land Grant University and a research-intensive University. As a Land Grant Institution, we are responsible for the Arizona Cooperative Extension System present in every county and directly charged with enhancing statewide economic development. It is critical for UA to provide a comprehensive Cyberinfrastructure (CI) plan and roadmap that facilitates collaboration and enables faculty, staff and students to conduct leading-edge research. The UA CI plan aligns with and takes inspiration from the 2007 NSF Cyberinfrastructure Vision for 21st Century Discovery report. As defined in the report, CI consists of computing systems, data storage systems, data repositories, specialized instruments, visualization environments, and people, all linked together by software and advanced networks to improve scholarly productivity and enable breakthroughs that would otherwise not be possible. UA cyberinfrastructure, policies, and training create an ecosystem of services to support research throughout its lifecycle.

The following committees provide research focus and strategy for implementing CI at UA: Research Computing Governance Committee (RCGC), and the Data Science Research and Training (DSRT) committee. RCGC has ad hoc subcommittees tasked with leading strategic advisory efforts: High Performance Computing (HPC) Technology Refresh, Data Management & Curation, and HPC Policies & Procedures. Subcommittee membership is intended to be transdisciplinary.

University of Arizona Research Cyberinfrastructure Goals

As research becomes increasingly complex, it requires more computational resources and associated training to enable researchers. The primary objective of the CI plan is to provide details on the necessary infrastructure, tools, technologies, training, and supporting expertise that will ensure that our researchers are able to accomplish their research goals and successfully collaborate with colleagues around the globe. Delivering the promise of modern research and digital literacy presents challenges for existing computing, network infrastructure, storage, and software tools. The goals outlined below are key to providing UA researchers with supported and sustainable cyberinfrastructure to meet current and future research needs.

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1. UA Research Cyberinfrastructure Ecosystem

The computing, networking, and storage demands of campus research are continuously expanding in scope and complexity. This presents a challenge to both central and discipline-specific infrastructure and support staff. Providing a more robust ecosystem of infrastructure, integrating central and distributed resources, and providing a coherent campus-wide cyberinfrastructure addresses these challenges. The CI ecosystem is developed and sustained through agreement and cooperation among myriad university organizations.

1.1 Research Technologies Cyberinfrastructure Resources

Last Updated: 2026-10-15

Point of Contact(s): Chris Reidy, chrisreidy@arizona.edu

The University Information Technology Services (UITS) team has maintained centrally funded and managed research computing infrastructure, infrastructure administration, and associated research consulting for High Performance Computing (HPC), Visualization, and Statistics since 1985. Recurring funding from UA has allowed for refresh of the centrally supported HPC systems every 3-5 years. The HPC resources are available, at no cost, to all campus researchers.

UA researchers have an option to fund additional HPC compute nodes and leverage the central infrastructure and system administration support, known as 'buy-in.' This allows researchers to obtain high-priority access to the purchased nodes, which are centrally managed as part of the larger system. Unused compute cycles that might be available from the buy-ins are available to other campus researchers, at a lower priority. Buy-in can improve grant competitiveness due to lower costs that must be included in the grant (i.e., hardware only, no operations costs); evidence of campus cost-sharing; and more positive funding agency review of cost-effective centrally administered facilities. The following systems are available (or soon available) at no cost to researchers:

- Ocelote general purpose cluster was purchased 2016 and discontinued in August 2026. However, the 95 Nvidia P100s were retained and will be integrated to Puma (no confirmed date yet).
- Puma general purpose cluster purchased 2020. The vendor is Penguin Computing. The system specifications are:
    - 48 Altus XE2242 CPU Nodes with 96 CPU x 4 = 384 CPU, 2.4GHz, 512GB RAM per node
    - 6 Altus XE2214GT GPU Nodes with 96 CPU, 2.4GHz, 512GB RAM, 4 32GB V100S per node
    - 2 Altus XE1212 High Memory Nodes with 96 CPU, 2.4GHz, 3072GB RAM per node
    - Total CPUs: 19,200 2.4 GHz AMD (EPYC 7642 Rome). Since the original purchase of Puma, researchers have added capacity with their funding that has taken the core count to 28952, which is a 50% increase. Those nodes are available for windfall when not in use by the acquiring PI.

- As of October 2026 Lynx is being installed and undergoing testing and integration. It is scheduled for general availability early 2027. The vendor is Lenovo. The system specifications are:
    - 40 ThinkSystem SR645 V3 Nodes with two AMD EPYC 9655 96 core 400W 2.6GHz Processors, 768 GB DDR5 RAM, 1.92TB NVMe, NDR200 Infiniband

- 2 ThinkSystem SR675 V3 Nodes each with 8 x Nvidia H200 GPUs, two AMD EPYC 9455 48 core 300W 3.1GHz processors, 1536GB DDR5 RAM, 1.92TB NVMe, NDR200 Infiniband
- 1 ThinkSystem SR645 V3 High Memory Node with two AMD EPYC 9455 48 core 300W 3.1GHz processors, 3072GB DDR5 RAM
- At installation time there are 25 Compute nodes, 38 GPUs, and one high memory node for researcher buy-in. The have the same specification as the centrally purchased nodes.
- Total CPUs: 7,962 AMD Turin cores from the central purchase. 5,376 cores from the initial round of buy-in.

Allocation. Each PI is allocated 150,000 CPU hours/month on the 2020 Puma cluster, with additional windfall processing time available on these systems. The allocations for Lynx have not yet been determined.

Facilitation. UITS has five full-time facilitators with expertise in HPC and data visualization (one position is open); and three full-time staff dedicated to UA HPC Systems Administration (one position is open). Research Computing Support facilitation includes software installation, support for scaling and benchmarking of parallel code, container technologies, and expertise in specialized accelerator technologies such as NVIDIA GPUs.

1.2. Storage

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Centrally administered HPC storage follows the same funding, support and upgrade models as the HPC computational resources. The UA HPC storage array is housed in the same research data center as the HPC.

The Qumulo P184 SSD based storage array was designed for reliability, performance, and scalability. To accommodate continued I/O demands in research data, this array provides 2PB of raw capacity, 65GB/s of read bandwidth and 44GB/s of write bandwidth. This is an all-flash solution which is very fast and very expensive. One of the benefits to researchers is that the computation times are not delayed by highly intensive IO operations from multiple users.

This storage array is mounted on both clusters and contains all of the software packages which are installed as modules, which means that the exact same applications can be run on any cluster.

Access to the HPC resources includes 50GB of home directory space, 500GB of space for each PI and an allocation of time-limited space up to 20TB available on request to PI's.

In late 2020 another array was provisioned that is available for rental. The rate is $47.35 per TB per year. It is mounted as /rental and directly to the HPC clusters. It is intended as the primary storage location for research data and not intended for running compute against. The all flash array is for that purpose.

An AWS solution is available that is more like archival. It uses AWS Intelligent Tiering between S3 and Glacier and is subsidized by UITS. Researchers will copy the data to the service and it will automatically migrate the data to Glacier and then to Deep Glacier.

1.3. Research Data Center Facilities

Last Updated: 2026-10-15
Point of Contact(s): Chris Reidy, chrisreidy@arizona.edu

UA has two facilities to house research computing resources:

1. Research Data Center (RDC): 1200 ft^2 raised floor data center designed for water-cooled racks dedicated to centrally managed research computing systems and large grant funded systems, with capacity for 40 standard racks. RDC capacity allows UITS to maintain three generations of centralized large computer clusters simultaneously, providing continuity of research while each new system is phased in.

2. Co-location Data Center: 1900 ft^2 of raised floor data center space for air-cooled research computing systems purchased and administered by colleges, departments, and research projects, with capacity for 70 standard racks.

Other than a nominal installation cost, no bandwidth or other recurring charges are levied for co-located research systems in these facilities.

UITS data centers are in the Computer Center with 1325 kW of battery backup and a 1750 kW generator for backup power. Cooling in the RDC is both in-rack cooling with chilled water heat exchangers and 70 tons of Computer Room Air Conditioning (CRAC) units. The UA Data Center is cooled with chilled water CRAC units. Both data centers are equipped with 18” raised floors that allow full coverage of cooling to all the equipment and equipped with leak detection systems in the subfloor. The fire suppression system is a multi-tiered defense with clean agent compressed gas, dry pipe pre-action sprinkler and EPO (Emergency Power Off) systems zoned to deploy in affected areas. UITS data centers have badge swipe access with two-factor authentication and video surveillance in the data centers and throughout the building. The data centers are monitored by a co-located 24/7 Operations staff and dedicated infrastructure team. Automated environmental and system monitoring assist with issue triaging and escalation. All personnel with access to unescorted data centers have undergone background checks and are required to be US Citizens.

1.4 Regulated Research Data and Computing Environments

Legal Aspects of Regulated and Controlled Data

Last Update:
Point of Contact: John Howard, HIPAA Compliance Office, jfhoward@email.arizona.edu

Regulated Research Environments (CUI, HIPAA, etc)

Last Updated: 2020-1-08
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The University of Arizona Information Technology Services department provides and manages secure research environments within the AWS GovCloud service, available to UArizona researchers to support security requirements for CUI and CUI / Export Controlled research projects. A small pilot will be launched Q1 2020 to support research using protected health information regulated by the Health Insurance Portability and Accountability Act of 1996 (HIPAA) resulting in a late spring / early summer 2020 service launch of the environment for projects under the provisions of a Data Use Agreement (DUA) but are not required to meet the additional requirements that pertain to data used as part of a standard of care practice. These environments will provide researchers computational and storage resources for active regulated data analysis. Management of the environment and security controls as well as technical support will be centrally funded.

Services provided:

- Compliant computational and storage resources for active regulated data analysis
- Project planning and budgeting
    - Consulting regarding fit and needs of project
    - Proper sizing of computation and storage resources
    - Analysis regarding the computational and storage resource costs associated with project
- Deployment and decommissioning of computational and storage resources
- Environment account access and management
- Technical support for resources provided by the service

1.5 ACCESS and National Artificial Intelligence Research Resource Pilot (NAIRR) National Computing Resources

Last Updated: 2026-10-15
Point of Contact(s): Chris Reidy (chrisreidy@arizona.edu)

If additional computing resources beyond UA HPC are needed, the ACCESS national supercomputer network and NAIRR are available to any United States-based researcher. These resources are freely available through an allocation request. See https://www.access-ci.org for more information. ACCESS has large supercomputers available with a variety of resources available while NAIRR is focused more on AI related resources

The UA Research Computing department is committed to helping UA researchers apply for allocations and generally support researchers leveraging these resources.

1.6 UITS Network Capacity

Last Updated:
Point of Contact(s): Dragana Vasic (dvasic@email.arizona.edu)

In addition to direct connections to commodity Internet service providers, UA is a member of Internet2 (I2). I2 is a not-for-profit organization that has built a 100Gb (gigabit) ultra-high speed backbone network for advancing Research and Education nationwide. The UA connection to I2

is through the Sun Corridor Network – a regional network established through a collaborative effort by Arizona State University (ASU), Northern Arizona University (NAU), and UA. The Sun Corridor Network provides advanced networking services beyond those available from the individual Arizona Universities and builds an environment essential to leading-edge education, research, and the sharing of digital communications resources, network services, and applications among eligible members. UITS manages and operates the Sun Corridor Network. The current connection from UA to Sun Corridor is a primary 100Gb circuit and a secondary 10Gb circuit via two(2) redundant edge routers, while Sun Corridor is connected to I2 via dual 100G connections in Tucson and Phoenix. The UA campus network consists of eight(8) redundant core routers in two core sites and eight(8) distribution routers in three(3) distribution locations. The core and distribution layer devices are interconnected at multi-100Gbps and most campus buildings are connected at 10Gbps. All buildings on campus are dually connected to campus distribution locations. Most customer ports are 1Gbps.

Researchers may also use a Science DMZ for fast high volume data transfers to outside collaborating institutions. The Science DMZ is deployed at the university network perimeter, outside border firewalls,via direct connections to a campus DMZ/Edge at multi-10Gbps. It is secured via static access lists deployed on two redundant, high throughput L3 switches dedicated to Science DMZ. There are two high-performance Data Transfer Nodes deployed in the Science DMZ managed by UITS.

The UA Research Data Center provides two(2) points of 40Gbps connections to the UA Data Center Fabric with all the servers connected by 1Gbps, 10Gbps, and 40Gbps connections. Servers are connected to two Cisco FEX switches for redundancy, as well as two Dell Z9500 high-density switches. Each pair of switching environments uplink to campus data center via four(4) 10Gbps circuits. This Data Center is located within the University network and is protected by both the campus border next-gen firewalls and by additional firewalls specific to enterprise and research Data Center services, providing higher-security for sensitive data than that available in high-speed Science DMZ.

The University of Arizona network is IPv6-capable with IPv6 routing enabled throughout the central network infrastructure. IPv6 service is enabled for departments and campus units upon request. The University of Arizona holds a /32 address allocation from ARIN and maintains native IPv6 connectivity to both Internet2 and the commodity Internet.

1.7 Data Science Institute (Data7). -> ORAI

Last Updated: 2019-11-15
Point of Contact(s): Nirav Merchant (nirav@email.arizona.edu), Maliaca Oxnam (maliaca@email.arizona.edu)

The UA Data Science Institute (Data7) was formed in 2017. Data7's goals include fostering the next generation of data-driven research by encouraging university-wide interdisciplinary collaboration, gaining external visibility, developing industry alliances, and increasing funding for UA research. By connecting UA researchers and aligning institutional expertise, computational resources, and infrastructure, Data7 enables investigators to ask more complex questions and achieve outcomes not easily attainable as individual investigators or within purely disciplinary teams. Data7 is also connected with the UA Transdisciplinary Research in Principles of Data Science (TRIPODS) effort, focused on the theoretical foundations of Data Science. Data7

actively coordinates and communicates training activities in cooperation with the Data Science Resources & Training (DSRT) group.

1.7.1 Project-based research support

Data7 solicits one-page white papers describing projects that fit the mission and goals of Data7 and fall into our focus areas:

- **Machine Learning**
- **Natural Language Processing**
- **Image Analysis**
- **Large-Scale Data Visualization**

Projects should combine medium- to large-scale, multidisciplinary research teams with the intention of pursuing external funding in six to 12 months. The purpose should be to produce proof of concept and results that provide competitive advantage when pursuing funding opportunities.

Pilot projects include Natural Language Processing for data collected from telephone conversations between cancer survivors and health coaches, medical diagnostic image classification using Machine Learning, wildlife camera image analysis, and support of work to classify transient events observed in galaxies.

1.7.2 Data Science Training

Data Science is an interdisciplinary endeavor, merging techniques from computer science, statistics, and math with domain-level concepts to increase discovery through data analytics and visualization. However, taking advantage of advanced computational resources and the "big data revolution" requires a set of specialized skills and core competencies.

Data7 works closely with the broader campus community to support a variety of learning activities. Examples include Data and Software Carpentry training, vendor-provided training (Cloud, Machine Learning, GPU etc.) in collaboration with UITS, and organizing special topic workshops (Containers, Workflow Managers). To foster Data Science Literacy, Data7 partners with the Data Science Research & Training (**DSRT**) group, developers of the Data Science Information Portal (**DSIP**), an online hub enumerating computational resources, expertise, and training events, and providing easy-to-use tools for UA researchers to find resources of interest. Data7 is also connected with the UA Transdisciplinary Research in Principles of Data Science (TRIPODS) effort, focused on the theoretical foundations of Data Science.

Data7 postdocs and technical staff in collaboration with DSRT and interested UA departments organize seminars and symposiums on Data Science-related topics. Data7 maintains an “Events” calendar to advertise training events and a “Classifieds” page that allows colleagues to to offer or ask for help with Data Science related issues.

1.8 Research, Innovation & Impact -> ORP

Last Updated: 2019-11-15
Point of Contact(s):

The **Office for Research, Innovation and Impact** (RII) supports the world-class research enterprise at the University of Arizona, which is ranked among the top 25 public universities nationwide with more than $687 million in research activity. Our researchers continue to forge innovative pathways, form powerful collaborations and make remarkable discoveries.

The UA research community is committed to preparing students to thrive in a time of augmented intelligence and the fusion of the digital, physical and biological worlds. Our research is creating the next generation of knowledge—and is translated and applied to society's grand challenges—to enhance economic development and promote a rich quality of life in Arizona and beyond.

Fundamentally, Research, Innovation and Impact is responsible for advancing transformative excellence in research across campus, with particular attention to our land-grant mission of service to the State of Arizona. We do so by enabling the research success of our faculty through supporting university research centers, institutes, museums and core facilities; providing research development, stewardship, compliance and safety services; and securing strategic external partnerships.

RII can help researchers with proposal development, finding funding, managing projects, and ensuring compliance with policies and procedures that promote the safe, legal, and ethical conduct of research. RII supports many Centers and Institutes and Core Facilities to support research. RII also provides the **Research Resources** gateway to information on research services, training & professional development, and core facilities, as well as links to forms & templates, policy & guidance and other useful information.

1.9 The Office of Digital Innovation & Stewardship (ODIS)

Last Updated: 2019-11-15
Point of Contact(s): Megan Finn Senseney (msenseney@email.arizona.edu)

The University of Arizona Libraries' Office of Digital Innovation and Stewardship supports the campus community in navigating a rapidly evolving digital landscape. We offer training and consultation on topics related to data-intensive research and digital scholarship, and we support open access and data sharing to advance global equity and the creation of new knowledge. Through partnerships and collaborations we preserve and provide access to distinctive digital collections and the scholarly record of the University of Arizona.

1.9.1 Research Data Repository

In an effort to foster open, reproducible, and collaborative research on campus through long-term stewardship, the Libraries are developing data repository services intended to provide responsible curation for research datasets and related supplementary research outputs (e.g., software/code, media) generated through research activities by University of Arizona-affiliated

researchers. Initial implementation is focused on establishing an institutional instance of the Figshare platform and developing associated data-related services for the storage and public sharing of research data, code, and related research products produced by University of Arizona researchers and affiliates. It also includes the associated data management and other services needed for ensuring quality and ease of use (e.g., data curation, and integrations with other campus systems). Our interest is sharing research products that can be made public, and the data repository is not meant as a general data store for private information, data related to the business functions of the University, or any kind of Regulated data. We are currently seeking early adopters with data ready for deposit; interested researchers should contact Fernando Rios (frios@email.arizona.edu).

1.9.2 Scholarly Communication and Research Information Management (SCRIM)

1.9.2.1 Scholarly Communication Services

The Libraries provides workshops and consultations on a range of issues in scholarly communication, including identity management, open access, fair use, and copyright. We offer assistance with ORCID and DOIs, and we also host a selection of scholarly journals edited and managed by UA students, faculty, and affiliates. For individual consultations, please contact Ellen Dubinsky (edubinsky@email.arizona.edu). Upcoming programming is listed at https://new.library.arizona.edu/events.

1.9.2.2 Campus Repository

On April 4, 2016, the University of Arizona Faculty Senate passed an open access policy that calls on the faculty and university to distribute faculty-authored scholarly articles to the widest possible audience through the UA Campus Repository. Our institutional repository facilitates open access to the research, creative works, publications and teaching materials of the University through open licensing. We collect, share, and archive content that has been selected and deposited by faculty, researchers, staff and affiliated contributors, and deposited content is discoverable via Google, Google Scholar, and other search engines. Eligible content includes pre-prints, departmental publications, technical reports, white papers, theses, and dissertations. Contact repository@u.library.arizona.edu with questions.

1.10 The CATalyst Studios

Last Updated: 11-25-2019
Point of Contact(s): Jennifer Nichols, (jtn@email.arizona.edu)

The CATalyst Studios is located in the Main Library as part of the new Student Success District. CATalyst Studios features a Data Studio, Virtual Reality Studio and Maker Studio. All facilities are open for use by all members of campus and the community. Learning and use of spaces is facilitated with workshops, user certifications and drop-in support.

1.10.1 Data Studio

The Data Studio features a 6x3 Data Visualization wall of Samsung UH46N-E displays, with a pixel count of 11520 x 3240.

1.10.2 VR Studio

The Virtual Reality Studio features the latest tethered and stand alone headsets, including Oculus Rift, HTC Vive, and Oculus Go, as well as stations for 3D modeling and a green screen cyclorama.

1.10.3 Maker Studio

The Maker Studio provides access to prototyping equipment for learning communities of all types. Technology includes 2 Epilog FusionPro laser cutters, Ultimaker 3 3D printers, Shapeoko CNC router, Roland Camm-1 Vinyl Cutter, Juki industrial sewing machine, arduino, raspberry pi. User needs regularly inform what technology and supplies are provided.

1.11 Advanced Visualization at the University of Arizona

Advanced data visualization:

- Data visualization wall
- VR/AR space

1.12 Health Sciences

The University of Arizona Health Sciences is the statewide leader in biomedical research and health professions training. The UArizona Health Sciences includes the Colleges of Medicine (Tucson and Phoenix), Nursing, Pharmacy, and the Mel and Enid Zuckerman College of Public Health, with main campus locations in Tucson and the growing Phoenix Biomedical Campus in downtown Phoenix. From these vantage points, the Health Sciences reaches across the state of Arizona and the greater Southwest to provide cutting-edge health education, research and community outreach services. A major economic engine, the Health Sciences employs nearly 5,000 people, has approximately 900 faculty members and garners $200 million in research grants and contracts annually.

The University of Arizona Health Sciences is a major contributor to the UA's well-earned reputation as one of our nation's top research institutions. From investigating biological concerns at the molecular level to studying and understanding individual communities, UAHS researchers are achieving breakthroughs that will improve individuals' lives. Numerous centers and institutes for specialized research and patient care function under the auspices of the UA Health Sciences.

The University of Arizona Center for Biomedical Informatics and Biostatistics (CB2) provides a wide variety of services and tools, as well as regularly scheduled educational sessions. The StatLab provides statistical and bioinformatics consulting.

1.12.1 Surveys and Clinical Databases

The University of Arizona Center for Biomedical Informatics and Biostatistics (CB2) is pleased to be a partner in the Research Electronic Data Capture (REDCap) Consortium. The REDCap consortium is composed of thousands of active institutional partners in over one hundred

countries who utilize and support REDCap in various ways. It supports a secure web application (REDCap) designed exclusively to support data capture for research studies and institutional projects. The REDCap application allows users to build and manage online surveys and databases quickly and securely. UA Health Sciences (UAHS) currently hosts hundreds of production and development REDCap projects. Periodic training on REDCAP is available.

2. Improving Institutional Efficiency

Commented [1]: Blake to update

Last Updated:
Point of Contact(s):

For the past 2 years, UA has been increasingly leveraging cloud services to reduce the need to maintain on premises servers. Key examples are: Student email services have been migrated to Google, Box online storage and collaboration tool has been licensed, UAccess Research Gateway launched in Amazon Web Services (AWS) in April 2015.

UA is home to many national and international research efforts that have developed significant CI dedicated for their projects and research communities, in collaboration with a global community of researchers from CISE (Computer Information Sciences and Engineering) disciplines these projects are continuously advancing CI capabilities. Presence of these projects, their active CISE collaboration and engagement with the campus infrastructure groups has helped guide the UA IT Strategic plan, incorporating some of the best practices and technologies from these projects to widely benefit UA researchers. This synergistic activity has ensured efficiency and informed decision-making as many of these projects rely on UA central IT capabilities and are able to scale their capabilities due to this active collaboration. Examples of upcoming activities include:

- Evaluation of a campus wide data management infrastructure using iRODS
- Utilizing Cloud computing based infrastructure to establish a statewide geospatial data sharing grid (AEGIS: Arizona Environmental Grid Infrastructure Service)

3. Developing Software Tools

Last Updated:
Point of Contact(s):

A crucial aspect of building a consolidated campus cyberinfrastructure is adopting and developing a common set of tools, applications, and middleware. These should be compatible with national cyberinfrastructure platforms ensuring robust and secure identity management with appropriate access and authorization capabilities, and integration with regional and national initiatives. To this effect UA has implemented its federated identity management (IdM) infrastructure using Shibboleth (Internet2) and supports enterprise wide single sign-on using Central Authentication Service (CAS); all campus Resources requiring IdM make use of this institutional infrastructure. Campus researchers are provided with technical support, sample source code and guidance to integrate their application with the institutional IdM. UA is an active member and contributor to both Internet2 and JASIG initiatives and UITS implements many software components developed by these organizations. We are part of national efforts such as Federated InCommon and CILogon. Relying on these enabling CI components, UA researchers are able to securely and reliably connect with multiple internal and external resources.

4. CI Support and Training

Last Updated:
Point of Contact(s):

An important focus for the CI plan is continuing education, training and outreach activities aligned to its CI vision. Currently these include for credit course work (graduate and undergraduate) offered through multiple departments and special topic workshops. In order to ensure successful and effective utilization of CI, researchers are provided access to technical support infrastructure that includes wikis, email lists and in-person consultation with technical staff. With greater access to distance education such as workshops provided through NSF XSEDE and vendor provided training, UA CI support and training effort will further incorporate these opportunities with in-house capabilities, with the goal of CI aligned workforce development. UA campus hosts regular workshops in collaboration with Software Carpentry and Data Carpentry foundations and projects such as CyVerse to educate researchers on best practices and establish a well informed community of practice on campus.

4.1 Data Science Resources & Training

Last Updated: 2019-11-15
Point of Contact(s): Jeff Oliver (jcoliver@email.arizona.edu), Vignesh Subbian (vsubbian@email.arizona.edu)

The rapidly expanding data science endeavor is quantitatively and qualitatively transforming research. Improved compute and storage, combined with novel approaches and algorithms provide researchers a number of tools for predictive analytics, classification, and visualization. This revolution is largely benefiting from the application of advances in statistics and computer sciences to specific domains in the social and natural sciences. Such approaches allow researchers to extract knowledge from large data sets and address previously unasked questions. However, a significant majority of students and faculty lack sufficient data science literacy skills to take advantage of the expanding landscape of data science tools.

Support for development in these skills and models is generally lacking in most formal training programs outside of computer sciences, despite a demand across the sciences for this training, especially in research computing. In many cases, these training demands are outstripping the currently accessible resources, creating a critical gap between the needs of scientists and their respective curricula. Not only does this knowledge and skills gap prevent researchers from applying data science approaches to their work, it limits the types of questions they even know to ask. Many researchers are restricted by this gap, yet there are more or less isolated pockets of data science expertise within each college. Researchers needing data science training would benefit from collaborations and connections with those data science practitioners in their respective domains.

The University of Arizona provides many computational resources and training opportunities in skills necessary critical for data science. The Data Science Resources & Training (DSRT) group was established in 2017 to ensure these opportunities are exploited by the broad audience of researchers at the University of Arizona. The DSRT steering committee advocates, advises, and cultivates new data science opportunities at the University of Arizona. Members of the steering committee connect the University of Arizona research community with data science training opportunities and campus resources by:

- Actively promoting data science resources within their respective professional communities on campus
- Promote training events and information on campus resources
- Collecting feedback and reporting back on training successes and gaps in opportunities
- Identifying additional data science trainers and opportunities

4.1.1 Data Science Ambassadors

Last Updated: 2019-11-15
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One specific effort of DSRT, the Data Science Ambassadors (DSA) program, was developed to help meet the growing demands for data science training across the University of Arizona campus. The program aimed to create accessible support within individual colleges by capitalizing on domain knowledge and data science expertise in the graduate student population. Graduate students' ability to "speak the same language" as researchers in their respective college reduces the communication barrier many researchers face when learning data science applications. The ultimate goal of the program is for graduate students to assist researchers and educators in navigating data science campus resources and to provide discipline-specific data science training.

The 2018-19 cohort of eight ambassadors engaged in a number of data science training events. Much of the efforts were through (1) Carpentries workshops: four of the ambassadors served as helpers for three Carpentry workshops and (2) "Astro Code" workshops organized an Ambassador. The three Carpentries workshops provided training to a combined 143 members of the UA campus. The five Astro Code workshops had an average attendance of 13 participants, and featured workshops on parallel processing and container technology. Two of the ambassadors were instrumental in organizing the Women in Data Science Tucson event on 5 April 2019. Over 100 people attended the event, which showcased numerous data science efforts on the UA campus. The 2019-2020 cohort of 10 ambassadors represents a growth in the number of colleges with an ambassador (six colleges, up from three). All but one of the ambassadors completed the Carpentries Instructor Training program as part of their ambassadorship.

Supported in part through funding from the Data Science Institute, the primary goal of the DSA program is to improve data science literacy at the UA campus. Beneficiaries include the college the ambassador serves and the ambassadors themselves. Colleges benefit from having a single point person to assist in training and navigation of campus data science resources. Actively countering the "silo" effect, ambassadors will create bridges among campus units, affording their respective colleges greater connectivity to the data science community. Ambassadors will immediately benefit from the training curriculum, as badged Carpentries instructors are increasingly recognized as top-quality candidates in job searches. The network of ambassadors and data science practitioners will also provide opportunities for ambassadors to improve their skills and understanding of data science applications.

4.1.2 Data Science Fellows

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The University of Arizona Health Sciences (UAHS) Office of Data Science Services (ODSS) Data Science Fellows initiative is starting in early 2020. The DSF program will consist of cohorts of postdocs with a mix of expertise in data science techniques (machine learning, image informatics, and natural language processing). Additionally, the DSF program will average six people per cohort and the Project Coordinator will coordinate the day-to-day operations and serve as the main point of contact for the scholars, program development team, mentor team, and other stakeholders involved in the program.
, and a makerspace with 3D printing.

5. Acknowledgments

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